

INTERNSHIP COMPLETION PRESENTATION

AgriVision — Machine Learning for Agribusiness

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Batch: 2024-28 | B.Tech CSE

YuvaIntern • Junior Machine Learning Engineer – Agribusiness

Organization & Role

YuvaIntern internship completed in the role of Junior Machine Learning Engineer – Agribusiness.

Internship Focus

Machine learning applications in agribusiness, including crop analytics, disease risk, data preparation, modeling and deployment.

Project Codename

AgriVision — a mobile-first ML solution designed to support early crop disease detection and agricultural decision-making.

Certificate Details

Certificate of Experience • Certificate No. YI/2026/208253/451552 • Issued October 08, 2026



The Problem

Smallholder farmers may depend on visual inspection and extension support for crop-disease diagnosis. Delays can allow disease to spread and can lead to yield loss or unnecessary agrochemical use.

The Proposed Solution

AgriVision uses crop images and machine learning to provide fast diagnostic support, with mobile access and low-connectivity alternatives such as SMS/USSD.

Expected Value

- Faster diagnosis
- Earlier intervention
- Better consistency
- Lower avoidable input waste
- Regional disease insights
- Scalable support for farmers

Core idea: shorten the gap between crop disease onset and effective treatment.

Defined the business challenge and target users.

Set SMART objectives for the pilot.

Designed a four-layer solution framework.

Created a 16-week phased roadmap.

Identified resources, roles, risks and evaluation criteria.

Pilot Targets

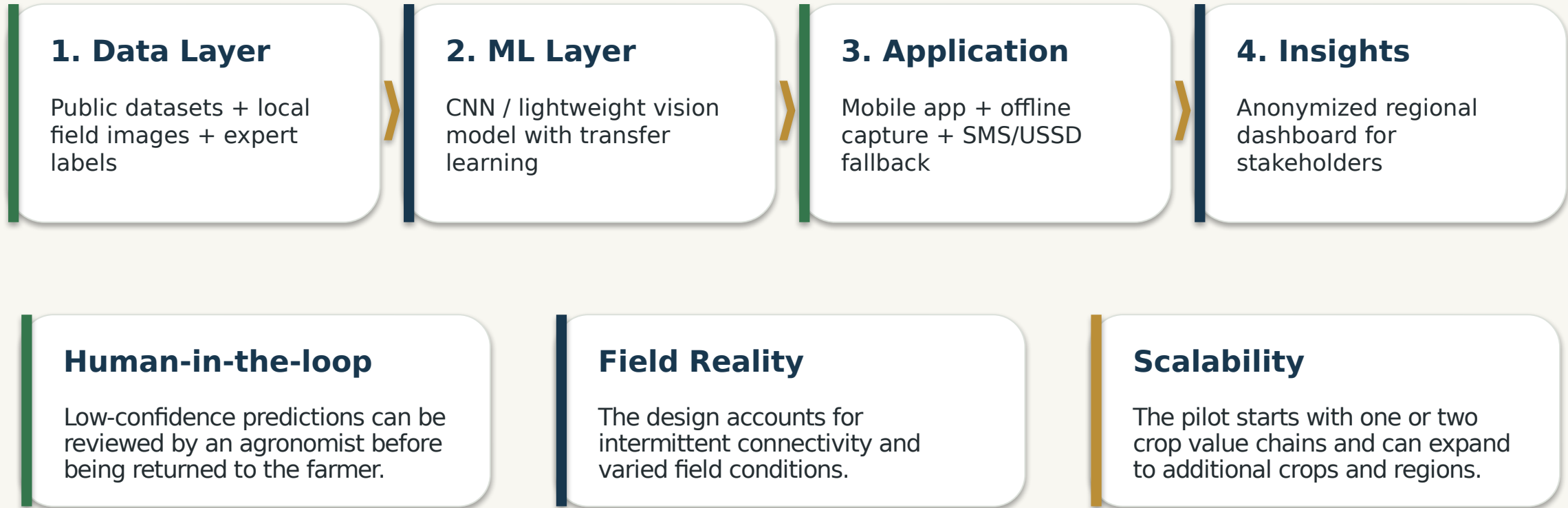
Top 5 target diseases • $\geq 90\%$ validation accuracy • diagnosis under 15 seconds

Diagnosis Goal

Reduce average turnaround from about 5–7 days to under 24 hours.

Architecture

Data layer → ML model → Mobile/SMS/USSD application → Regional insights dashboard



Agribusiness Data

- Crop yield records
- Weather time-series
- Soil quality measurements
- Market & price data
- Remote sensing / NDVI

The report emphasizes that agricultural data is seasonal, spatial and operationally messy.

Cleaning Methods

- Missing-value handling using time-/neighbour-aware methods
- Region/crop-specific outlier checks
- Normalization and standardization
- Log transformation for skewed variables
- Discretization / binning where useful

Why Domain Knowledge Matters

A genuine drought or heatwave should not automatically be removed as an outlier. Preprocessing must distinguish data errors from real agronomic variation.

Crop

Historical yield trend •
crop rotation • input
intensity

Weather

GDD • rainfall patterns •
extreme-event counts •
lag windows

Soil

N:P:K ratios • fertility
score • soil-crop
suitability

Cross-Dataset

Water balance • NDVI
trajectory • climate-
adjusted potential

Feature Selection

Filter → domain review → embedded
importance → stability check

Target Applications

Yield prediction • irrigation optimization •
price forecasting • disease/risk classification

Quality Checks

Consistent units • no data leakage
• validation across seasons/regions

Model Workflow

1. Train / validation / test split
2. Preprocessing
3. Baseline model
4. Candidate model training
5. Cross-validation
6. Model selection
7. Final test evaluation

Algorithm Choices

- Gradient Boosting — strong tabular accuracy
- Random Forest — robust fallback
- CNN + transfer learning — image-based disease detection
- SVM — useful for small-sample classification
- Time-series models — seasonal forecasting

Evaluation

Classification: Accuracy, Precision, Recall, F1, ROC-AUC, Confusion Matrix

Regression: MAE, RMSE, MAPE, R^2

Agricultural impact: economic loss avoided, actionability lead time, adoption readiness.

Deployment

Hybrid cloud-edge approach

- Containerized model
- Managed inference endpoint
- Lightweight mobile/PWA access
- Offline caching
- SMS/USSD fallback
- Staging → pilot → production

Integration & Scale

- Farm-management systems
- IoT / weather / soil feeds
- ERP/business systems
- REST/JSON APIs or message queues
- Versioned models
- Geographic and seasonal scaling

Communication

Farmers: simple local-language guidance

Agronomists: practical explanations + override

Managers: ROI, adoption and operations

IT teams: architecture, runbooks and SLAs

What to Monitor

- Model accuracy / error
- Data drift
- API latency & uptime
- Sync failures
- Adoption / usage
- Yield, cost and decision-time impact

Continuous Improvement

Feedback → ground-truth outcomes
→ drift checks → retraining →
staging validation → canary release
→ wider rollout

Key Risks & Mitigation

Low connectivity → offline +
SMS/USSD

Low trust → local champions + co-
design

Model drift → scheduled monitoring

Privacy → consent + governance

Wrong recommendations →
agronomist review



THANK YOU

AgriVision — Machine Learning for Smarter Agribusiness

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